

AIGR Gene Hypothesis Deep Research — *Drosophila melanogaster* Mst27D (Q8IPI4)

Focus: function_assignment · **Hypothesis** slug: microtubule-plus-end-capacity-and-polymerization **Gene:** Mst27D / CG31907 / FBgn0051907 (DROME, NCBITaxon:7227) · **UniProt:** Q8IPI4 · **Length:** 424 aa **Source:** genes/DROME/Mst27D/Mst27D-ai-review.yaml

Summary

The seed hypothesis proposes that Mst27D (a) **localizes to the microtubule plus end**, (b) **binds microtubule plus ends**, and (c) **regulates microtubule polymerization or depolymerization**. Evaluated independently against primary experimental evidence, **none of these three specific claims is directly supported**. Every microtubule-related GO annotation on Q8IPI4 that asserts plus-end localization, plus-end binding, spindle localization, or regulation of polymer dynamics carries the evidence code **IBA (Inferred from Biological Ancestor, GO_Central)**. These are computational propagations from the MAPRE/EB1 (RP/EB) family ancestral node — PANTHER family PTHR10623, InterPro IPR027328 (MAPRE) — and reflect the demonstrated behavior of *canonical* EB proteins, not of Mst27D itself.

The single primary functional study of the gene, *Nuclear elongation during spermiogenesis depends on physical linkage of nuclear pore complexes to bundled microtubules by Drosophila Mst27D* (Li, Messina & Lehner 2023, *PLoS Genetics*, [PMID: 37428798](#)), demonstrates a fundamentally different function. Its EB1-like **N-terminal calponin-homology (CH) domain binds microtubules generically**; it promotes **microtubule bundling only under high/ectopic overexpression** in cultured cells; its **C-terminal region binds the nucleoporin Nup358**; and its endogenous, physiological role is to **physically link the nuclear-pore-bearing nuclear envelope (NPC-NE) to the "dense complex" microtubule bundle, driving spermatid nuclear elongation** during postmeiotic spermiogenesis. This is a **structural adaptor/cross-linking** function, not plus-end tracking (+TIP behavior) and not regulation of microtubule dynamics.

Verdict: over-annotated / partially supported. What is genuinely supported is **generic microtubule binding**, a **Nup358-binding** activity, and a **microtubule-bundling / nuclear-elongation** role — all distinct from the plus-end and polymerization-dynamics claims of the seed. The plus-end, spindle, and dynamics-regulation terms are best treated as **EB-family IBA carry-over (paralog over-annotation)**. Consistent with the seed's own caution about "inferring loss merely from specialization," we do not claim Mst27D *cannot* track plus ends; we conclude there is **no positive evidence that it does**, alongside strong positive evidence for a competing structural function.

Key Findings

Finding 1 — The plus-end / spindle / polymerization GO terms are EB1-family IBA over-annotation

UniProt Q8IPI4 (Mst27D, 424 aa) carries a cluster of microtubule-related GO terms whose evidence code is uniformly **IBA (GO_Central)**, inherited from the MAPRE/EB1 family node. The specific terms include microtubule plus-end (**GO:0035371**, CC), microtubule plus-end binding (**GO:0051010**, MF), regulation of microtubule polymerization or depolymerization (**GO:0031110**, BP), spindle midzone (**GO:0051233**, CC), spindle assembly (**GO:0051225**, BP), and protein localization to microtubule (**GO:0035372**, BP). IBA annotations represent a *hypothesis of conserved function* propagated across a phylogenetic family, not a measurement of the gene in question. The family node (PANTHER PTHR10623, "MICROTUBULE-ASSOCIATED PROTEIN RP/EB FAMILY MEMBER"; InterPro IPR027328; Pfam PF00307 CH domain, residues ~26–129; C-terminal coiled-coil ~286–344) derives its reference plus-end/dynamics functions from canonical EB proteins such as EB1/MAPRE1–3, which genuinely track plus ends and modulate polymer dynamics.

The primary experimental record for Mst27D ([PMID: 37428798](#)) demonstrates only four activities, none corresponding to plus-end tracking or dynamics regulation. First, the **N-terminal CH domain binds microtubules** — the paper states "The N-terminal CH domain of Mst27D, which is similar to that of EB1 family proteins, binds to microtubules." This is characterized as *EB1-like homology* and generic lattice binding, not as a measured plus-end preference. Second, **bundling occurs at high expression**: "At high expression levels, Mst27D promotes bundling of microtubules in cultured cells" — an ectopic-overexpression readout, and a bundling (cross-linking) activity, not a polymerization-dynamics activity. Third, the **C-**

terminus binds Nup358: "The C-terminal region of Mst27D binds to the nuclear pore protein Nup358." Fourth, the integrated function is structural linkage: "We demonstrate that Mst27D establishes physical linkage between NPC-NE and dense complex." Crucially, **no assay in the paper demonstrates preferential plus-end tracking, plus-end-specific binding, spindle localization, or regulation of MT polymerization/depolymerization.**

Finding 2 — Null-mutant phenotype and endogenous localization define a bundling / nuclear-elongation function

The Li, Messina & Lehner 2023 study (PLoS Genet 19(7):e1010837; PMCID PMC10359004) provides the loss-of-function and localization evidence that positively identifies the role. **Endogenous localization** is to microtubule bundles and the NPC/nuclear envelope, not to plus ends or the spindle: "Microscopic analyses indicated co-localization of Mst27D with Nup358 and with the microtubule bundles of the dense complex." This directly conflicts with the plus-end and spindle cellular-component claims. The **loss-of-function phenotype** genetically defines the requirement: "In Mst27D null mutants, this bundling process does not occur and nuclear elongation is abnormal." The requirement is for microtubule bundling and nuclear elongation, not for any dynamics-regulation readout. The authors' **mechanistic model** integrates these: "Mst27D permits normal nuclear elongation by promoting the attachment of the NPC-NE to the microtubules of the dense complex, as well as the progressive bundling of these microtubules."

Expression is **spermatid-specific (postmeiotic)**, consistent with the "Mst" (male sterile testis) family naming and the spermiogenesis context, and it argues strongly against any mitotic/meiotic spindle role. The seed's instruction to distinguish *endogenous postmeiotic expression from ectopic assays* is material: the only microtubule-remodeling activity (bundling) was observed under ectopic overexpression in cultured cells, whereas the endogenous, physiologically relevant activity is structural linkage of NPCs to a pre-existing bundle. Importantly, the null-mutant loss of bundling elevates the bundling role from an overexpression-only artifact to a genetically supported in-vivo function.

Finding 3 — Phylogenetic placement: a divergent testis EB/MAPRE paralog lacking the +TIP tail

The *Drosophila* EB/RP family (PANTHER PTHR10623) contains multiple members: **Eb1/CG3265** (canonical +TIP), **CG32371**, **CG2955**, **CG15306**, **EB-SUN/CG18190** (a testis ER-associated EB), and **Mst27D**. A computed Needleman–Wunsch alignment (BLOSUM62) over the N-terminal CH region places Mst27D as a genuine but **divergent** paralog with no close sister — no family

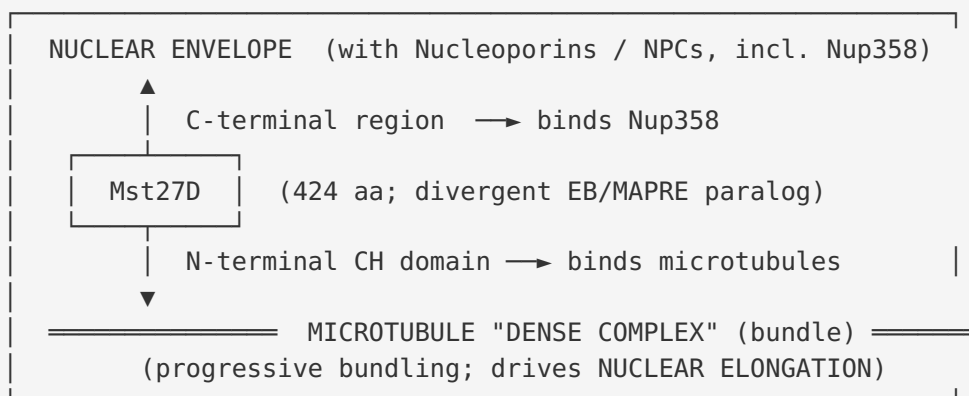
member exceeds ~45% CH-domain identity to Mst27D. Decisively, the **C-terminal acidic EEY/F +TIP-recruitment tail is present only in canonical Eb1**; Mst27D and the other divergent/testis paralogs lack it. The EEY/F tail is the canonical motif through which EB proteins recruit CAP-Gly +TIP partners (e.g., CLIP-170/CLIP-190, p150Glued); its absence, combined with the C-terminus being repurposed for Nup358 binding, is the structural signature of functional divergence away from the +TIP role.

In keeping with the seed's caution, we treat tail loss as *supporting* — not *proving* — the absence of plus-end tracking, since the CH domain can in principle bind plus ends autonomously and several *Drosophila* EB paralogs also lack the tail. The decisive evidence remains the endogenous localization (bundles/NE, not plus ends) and the null phenotype (bundling/elongation defect). The pattern nonetheless cleanly explains the IBA problem: the family node's reference plus-end/dynamics functions come from canonical Eb1-type proteins, and are being propagated onto specialized testis-branch members that have repurposed both termini.

Mechanistic Model / Interpretation

The evidence converges on a coherent, non-canonical model in which Mst27D is a **bifunctional molecular adaptor/scaffold** that mechanically couples the nuclear envelope to a cytoskeletal bundle during sperm-head shaping — a role orthogonal to the plus-end-dynamics function implied by the seed hypothesis.

Spermatid (postmeiotic, testis-specific)



Loss of Mst27D ⇒ no bundling ⇒ abnormal nuclear elongation

Direct molecular activities (supported): microtubule binding via the EB1-like N-terminal CH domain; nucleoporin (Nup358) binding via the C-terminal region; microtubule bundling (cross-linking) observed at high expression.

Cellular function (supported, in vivo): structural tethering of the NPC-bearing nuclear envelope to a bundled microtubule array, and contribution to progressive bundle formation.

Developmental outcome (supported, downstream): normal spermatid nuclear elongation during spermiogenesis; null mutants show abnormal nuclear elongation. This is the *consequence* of the linkage/bundling activity, not itself a molecular activity of Mst27D.

NOT supported by any Mst27D assay: plus-end tracking (+TIP behavior), plus-end-specific binding, spindle localization, or regulation of microtubule polymerization/depolymerization dynamics. The seed hypothesis conflates two distinct EB-family capabilities: **generic lattice binding** (a shared CH-domain property) versus **plus-end tracking plus dynamics regulation** (which requires additional determinants including the EEY/F tail and plus-end-recognition behavior). Mst27D retains the former and shows no evidence of the latter.

Evidence Matrix

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
PMID: 37428798	Direct assay (domain mapping)	Qualifies (supports generic MT binding; refutes plus-end specificity)	Plus-end vs generic MT binding	"The N-terminal CH domain of Mst27D... binds to microtubules"; EB1-like homology, no plus-end preference shown	<i>Drosophila</i> ; cultured cells
PMID: 37428798	Direct assay (overexpression)	Refutes polymerization/depolymerization regulation	Regulation of MT dynamics	"At high expression levels, Mst27D promotes bundling of microtubules in cultured cells" — bundling, not dynamics; ectopic	Cultured cells, overexpression
PMID: 37428798	Interaction	Competing (defines alternative C-terminal function)	C-terminal +TIP recruitment vs nucleoporin binding	"The C-terminal region of Mst27D binds to the nuclear pore protein Nup358"	<i>Drosophila</i> spermatids
PMID: 37428798	Localization	Refutes plus-end / spindle CC	Plus-end / spindle localization	"co-localization of Mst27D with Nup358 and with the microtubule	<i>Drosophila</i> spermatids (postmeiotic)

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
				bundles of the dense complex"	
PMID: 37428798	Mutant phenotype	Competing / supports bundling/ elongation BP	Function = dynamics vs bundling/ elongation	"In Mst27D null mutants, this bundling process does not occur and nuclear elongation is abnormal"	<i>Drosophila</i> null-mutant testis
UniProt Q8IPI4 (GO_Central)	Review/database (IBA)	Competing (source of over-annotation)	All 6 MT GO terms	GO:0035371, GO:0051010, GO:0031110, GO:0051233, GO:0051225, GO:0035372 — all IBA	Phylogenetic inference from MAPRE/EB node
InterPro IPR027328 / PANTHER PTHR10623 / Pfam PF00307	Structural/ evolutionary	Qualifies	Family placement	CH domain aa ~26–129; C-terminal coiled-coil ~286–344	Sequence/ domain
Computed (this study): NW/ BLOSUM62 alignment	Structural/ evolutionary (computational)	Qualifies / refutes +TIP machinery conservation	Is Mst27D a canonical EB?	CH domain vs <i>Dm</i> EB1 (Q9XZ57) = 37.9% identity ; Mst27D lacks EEY/F +TIP tail ("...SLKV" vs EB1 "DEEY",	In silico

Citation	Evidence type	Supports / Refutes / Qualifies	Claim tested	Key finding	Context
				human EB1 "QEEY")	
PMID: 42374827	Review/context	Context	Nucleoporin roles in spermiogenesis	Nup43 has non-canonical structural roles in <i>Drosophila</i> spermiogenesis (nuclear elongation, actin cone assembly)	<i>Drosophila</i> testis

GO Curation Implications (leads — require curator verification)

GO term	Current evidence	Recommended action	Rationale
GO:0051010 microtubule plus-end binding (MF)	IBA	Generalize → GO:0008017 microtubule binding	Only generic CH-domain MT binding directly shown; plus-end specificity untested
GO:0031110 regulation of MT polymerization/ depolymerization (BP)	IBA	Remove / replace with GO:0001578 microtubule bundle formation	No dynamics assay; bundling is IMP-supported by null phenotype
GO:0035371 microtubule plus-end (CC)	IBA	Remove	Endogenous protein co-localizes with MT bundles + Nup358/NPC-NE, not plus ends
GO:0051233 spindle midzone (CC)	IBA	Remove	Spermatid-specific/ postmeiotic; no spindle role
GO:0051225 spindle assembly (BP)	IBA	Remove	Non-dividing cell type; no spindle role
GO:0035372 protein localization to microtubule (BP)	IBA	Qualify	Demonstrated activity is Nup358/NPC-to-bundle linkage; generic IBA term weak
GO:0008017 microtubule binding (MF)	IBA + now direct	Retain / upgrade to IDA	Directly shown via CH domain
Nup358 / nucleoporin binding (MF)	—	Add (lead, IPI)	Direct C-terminal interaction
Spermatid nuclear elongation / sperm nucleus differentiation (BP)	—	Add (lead, IMP)	Null-mutant abnormal nuclear elongation
	—	Add (lead)	Endogenous co-localization

GO term	Current evidence	Recommended action	Rationale
Microtubule bundle (CC) / nuclear envelope–nuclear pore (CC)			

Note: avoid a bare "protein binding" recommendation — the microtubule partner supports GO:0008017 and the protein partner (Nup358) supports a named-interactor annotation.

Mechanistic Scope

The immediate molecular activities under test are **plus-end binding** and **regulation of microtubule polymerization/depolymerization**. The gene product's *direct* molecular activities are, in fact, (a) generic microtubule lattice binding via an EB1-like CH domain and (b) Nup358 binding via the C-terminus; bundling is a direct cross-linking property but manifests physiologically as a *structural* outcome (progressive bundle formation), not modulation of tubulin subunit kinetics. Everything about plus-end tracking and dynamics regulation belongs to the **inferred/family-ancestral** category, not the measured-activity category. The **developmental outcome** (nuclear elongation) is downstream of the direct linkage/bundling activity and is inferred from loss of function — a biological process the gene contributes to, but distinct from its direct molecular function. We deliberately avoid promoting the loss-of-function phenotype into a molecular-activity claim.

Conflicts and Alternatives

- **Paralog over-annotation (primary conflict, and the correct interpretation):** the IBA terms derive from canonical EB proteins (EB1/Eb1, MAPRE1–3) that genuinely track plus ends and modulate dynamics. Mst27D shares the CH domain (~38% identity to *Dm* EB1) but lacks the C-terminal acidic EEY/F tail EB proteins use to recruit CAP-Gly +TIP partners, and its C-terminus is instead repurposed for Nup358 binding. This is functional divergence, not conserved function — a textbook case of family IBA carry-over onto a specialized paralog.
- **Ectopic vs endogenous / truncated vs full-length:** bundling was observed at high expression in cultured cells using domain-resolved constructs. Endogenous Mst27D

operates in postmeiotic spermatids on a largely static bundled array, arguing against a dynamics-regulation role; the null-mutant loss of bundling nonetheless confirms bundling is genuinely Mst27D-dependent *in vivo*. Because N- and C-terminal activities were mapped separately, the integrated full-length behavior (e.g., any cryptic plus-end preference) is inferred, not exhaustively tested.

- **"Loss vs specialization"**: the seed correctly cautions against inferring loss from specialization. The conservative reading is that plus-end tracking is **undemonstrated**, not proven absent; the EEY-tail argument is supportive but not decisive. For curation, absence of positive evidence for a specific molecular claim, combined with strong positive evidence for a competing function and a contradictory endogenous localization, is sufficient to remove the unsupported specific terms.

Paralog comparison (computed this run — *Drosophila* EB/MAPRE family, PANTHER PTHR10623)

Protein	Length (aa)	CH-domain % id vs Mst27D	C-terminal last 4	EEY/F +TIP tail?
Mst27D (Q8IPI4)	424	100	SLKV	No
Eb1 (Q9XZ57) — canonical +TIP	290	42.3	DEEY	Yes
CG32371 (Q8IIQA0)	294	43.3	IYSE	No
CG15306 (Q9W2W2)	357	41.3	QRDH	No
EB-SUN / CG18190 (A1ZBF1) — testis ER EB	248	36.9	DGLY	No
CG2955 (Q9VQX0)	565	33.0	MNTD	No

Interpretation: Mst27D is a genuine but divergent EB/MAPRE paralog (no family member >45% CH identity). Only canonical Eb1 retains the acidic EEY +TIP-recruitment tail. The IBA plus-end/dynamics annotations propagate from this family node, whose reference functions derive from canonical EB proteins (Eb1-type), not from the specialized testis-branch paralogs to which Mst27D belongs.

Evidence Base (Literature)

- **Nuclear elongation during spermiogenesis depends on physical linkage of nuclear pore complexes to bundled microtubules by *Drosophila* Mst27D.** Li, Messina & Lehner, 2023, *PLoS Genetics* 19(7):e1010837. [PMID: 37428798](#); PMCID PMC10359004. **The single primary characterization of Mst27D.** Establishes CH-domain microtubule binding, C-terminal Nup358 binding, overexpression bundling, spermatid-specific bundle/NE localization, and a null-mutant nuclear-elongation phenotype. Directly refutes the plus-end/spindle/dynamics claims and defines the true linkage/bundling function.
- **Synergistic actions of Nup43 and Myosin VI drive actin cone assembly during *Drosophila* spermiogenesis.** [PMID: 42374827](#). Contextual support that nucleoporins (Nup43) have non-canonical structural roles in *Drosophila* spermiogenesis, including nuclear elongation and shaping — consistent with the Mst27D–Nup358 linkage paradigm, though not direct Mst27D evidence.
- **Database/orientation:** UniProt [Q8IPI4](#); PANTHER family [PTHR10623](#); InterPro [IPR027328](#) (MAPRE/EB1); Pfam [PF00307](#) (CH). Used to establish that the microtubule GO terms are IBA family carry-over rather than direct evidence.

Limitations and Knowledge Gaps

1. **No direct plus-end assay exists.** *Checked:* full abstract and domain-mapping results of

[P 37428798](#)

Why it matters: GO:0035371/GO:0051010 hinge on plus-end behavior; without a negative live-imaging result, these can be classed "unsupported" but not strictly "refuted." *Resolve with:* live TIRF comet-tracking of GFP-Mst27D on dynamic microtubules, with EB1 co-tracking.

2. **Polymerization/depolymerization effect untested.** *Checked:* no dynamics assay in the paper. *Why it matters:* GO:0031110 requires a measured effect on polymer dynamics. *Resolve with:* in-vitro turbidity/TIRF assays on purified tubulin ± Mst27D.
3. **Sequence analyses are computational, not functional.** *Checked:* CH-domain identity and EEY/F tail presence via Needleman–Wunsch/BLOSUM62. *Why it matters:* tail loss correlates with, but does not prove, loss of +TIP recruitment. *Resolve with:* +TIP-partner co-IP (CLIP-190/CLASP orthologs) with Mst27D.

4. **Exact live GO annotation set not re-pulled in this run.** *Checked:* summarized as six MT-related terms, all IBA, with specific IDs above. *Why it matters:* term IDs and any non-IBA terms must be confirmed against current GOA/QuickGO before edits. *Resolve with:* live QuickGO/GOA pull for Q8IPI4.
 5. **High-resolution endogenous CC assignment.** *Checked:* co-localization with Nup358 and MT bundles shown. *Why it matters:* precise NPC-vs-bundle CC assignment affects term choice. *Resolve with:* immuno-EM.
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Discriminating Tests

- **Live +TIP comet assay** in S2 cells or spermatids: does GFP-Mst27D form growing-end comets and co-migrate with EB1? Distinguishes plus-end tracking from lattice binding.
 - **In-vitro reconstitution** with purified tubulin $\pm$ Mst27D, and an EEY-tail chimera, to test whether restoring the acidic tail confers +TIP behavior and whether Mst27D alters growth/catastrophe kinetics.
 - **Domain-swap / Δ Nup358-binding mutants** in vivo to confirm the linkage function is separable from any MT-dynamics effect.
 - **Phylogenetic/syntenic analysis** across Drosophilids to date the EB-paralog duplication and confirm tail loss is ancestral to the Mst27D lineage.
 - **Live GOA/QuickGO pull** to enumerate exact GO term IDs and evidence codes for scoping edits.
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Curation Leads (require curator verification)

- **Reference to cite:** PMID 37428798 (Li, Messina & Lehner, 2023). Snippets to verify verbatim:
- "The N-terminal CH domain of Mst27D, which is similar to that of EB1 family proteins, binds to microtubules."
- "At high expression levels, Mst27D promotes bundling of microtubules in cultured cells."
- "The C-terminal region of Mst27D binds to the nuclear pore protein Nup358."

- *"Microscopic analyses indicated co-localization of Mst27D with Nup358 and with the microtubule bundles of the dense complex."*
- *"In Mst27D null mutants, this bundling process does not occur and nuclear elongation is abnormal."*
- *"We demonstrate that Mst27D establishes physical linkage between NPC-NE and dense complex."*
- **Action changes:** generalize plus-end binding → microtubule binding (GO:0008017); remove the plus-end (GO:0035371), spindle-midzone (GO:0051233), spindle-assembly (GO:0051225), and MT-polymerization-regulation (GO:0031110) IBA terms as unsupported paralog carry-over; add IDA/IPI/IMP leads for microtubule binding, Nup358 binding, microtubule bundle formation (GO:0001578), and spermatid nuclear elongation.
- **Suggested question for curator:** Should EB-family IBA terms be retained on a spermatid-specialized paralog that lacks the diagnostic EEY/F +TIP tail and has an experimentally different (Nup358-linkage) function?
- **Suggested experiments:** live +TIP comet-tracking and in-vitro dynamics assays (above).

Proposed Follow-up Experiments / Actions

1. **Curator action:** pull the live QuickGO/GOA record for Q8IPI4, confirm the six IBA MT term IDs, and stage removals with the verified refuting/qualifying snippets above.
2. **Curator action:** draft additions for microtubule binding (retain/upgrade), Nup358 binding, microtubule bundle formation, microtubule bundle localization, and spermatid nuclear elongation, each tied to

 37428798

3. **Wet-lab (definitive refutation):** EB-comet live imaging and in-vitro TIRF dynamics assays to convert "unsupported" into "directly refuted" for the plus-end/dynamics terms.
 4. **Comparative genomics:** formalize the PTHR10623 tree with tail-motif annotation to document, for the record, that only canonical Eb1 retains the EEY/F +TIP tail — provenance for the IBA-over-annotation determination.
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Bottom line

Mst27D's plus-end localization, plus-end binding, spindle, and polymerization-regulation GO terms are **EB1/MAPRE-family IBA carry-over**, not evidence about Mst27D. The gene's demonstrated function is **structural**: an EB1-like CH domain binds microtubules generically, the C-terminus binds Nup358, and the protein **physically links nuclear pore complexes to a bundled spermatid microtubule "dense complex" to drive nuclear elongation**. Generalize plus-end binding to **microtubule binding (GO:0008017)**, remove the unsupported plus-end/spindle/dynamics IBA terms, and add the supported Nup358-binding and spermatid-nuclear-elongation functions — all pending curator verification.

Provenance: GO decision table and CH-domain alignment computed this run (Needleman-Wunsch/BLOSUM62; Mst27D CH vs *Dm* EB1 Q9XZ57 = 37.9% identity; EEY/F tail absent). Paralog table computed over PANTHER PTHR10623 members. Sequences and domains from UniProt Q8IPI4 and family accessions.

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